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|----------------|--|---|
| 8 (a) | Q/V , with symbols explained [do not allow in terms of units] | B1 [1] |
| (b) (i) | on a capacitor, there is charge separation/there are + and - charges
<u>either</u> to separate charges, work must be done
<u>or</u> energy released when charges 'come together' | M1
A1 [2] |
| (ii) | <u>either</u> energy = $\frac{1}{2}CV^2$ <u>or</u> energy = $\frac{1}{2}QV$ and $C = Q/V$
change = $\frac{1}{2} \times 1200 \times 10^{-6} (50^2 - 15^2)$
change = 1.4 J (1.37)
[allow 2 marks for $\frac{1}{2}C(\Delta V)^2$, giving energy = 0.74 J] | C1
C1
A1 [3] |